GCSE PHYSICS

Higher Tier – Paper 1 style

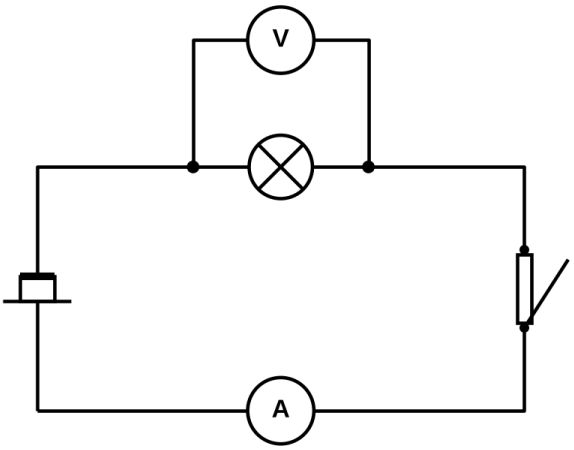
Topic 4.2: Electricity

Practice questions in the style of AQA GCSE Physics (8463). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 180 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **173**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 173

0 1 The diagram shows an electric circuit. The ammeter reads 0.40 A and the voltmeter reads 3.0 V.



0 1 . 1 Which component in the circuit measures current? **[1 mark]**

0 1 . 2 How is the voltmeter connected to the lamp? **[1 mark]**

0 1 . 3 Write down the equation that links charge flow, current and time. **[1 mark]**

0 1 . 4 The circuit is switched on for 2.0 minutes. Calculate the charge that flows through the lamp. **[3 marks]**

Answer = _____ C

0 1 . 5 Calculate the resistance of the lamp. **[3 marks]**

Answer = _____ Ω

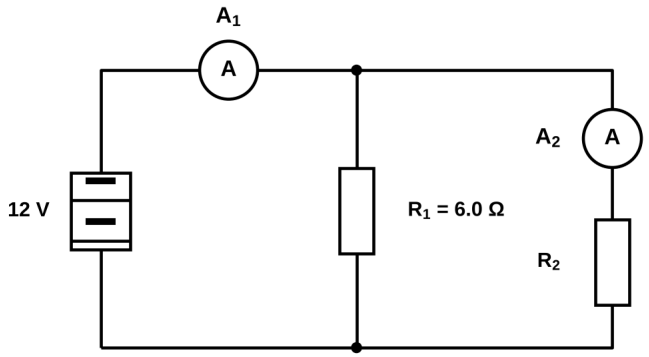
0 1 . 6 Calculate the energy transferred by the lamp in 2.0 minutes. **[3 marks]**

Answer = _____ J

0 1 . 7 The student adds a second identical lamp in series with the first. State what happens to the current in the circuit. **[1 mark]**

_____

0 2 The diagram shows a circuit with two resistors connected in parallel to a 12 V battery. Resistor R_1 has a resistance of 6.0 Ω.



0 2 . 1 Calculate the current through R_1. **[3 marks]**

Answer = _____ A

0 2 . 2 Ammeter A_2 reads 1.0 A. Calculate the reading on ammeter A_1. **[2 marks]**

Answer = _____ A

0 2 . 3 Calculate the resistance of R_2. **[3 marks]**

Answer = _____ Ω

0 2 . 4 Calculate the total resistance of the circuit. **[3 marks]**

Answer = _____ Ω

0 2 . 5 Explain why the total resistance is smaller than the resistance of R_1. **[2 marks]**

0 2 . 6 State the potential difference across R_2. **[1 mark]**

0 3 Two resistors of 4.0 Ω and 6.0 Ω are connected in series to a 5.0 V cell.

0 3 . 1 Calculate the total resistance of the two resistors. **[2 marks]**

Answer = _____ Ω

0 3 . 2 Calculate the current in the circuit. **[3 marks]**

Answer = _____ A

0 3 . 3 Calculate the potential difference across the 6.0 Ω resistor. **[3 marks]**

Answer = _____ V

0 3 . 4 State the potential difference across the 4.0 Ω resistor. **[1 mark]**

_____

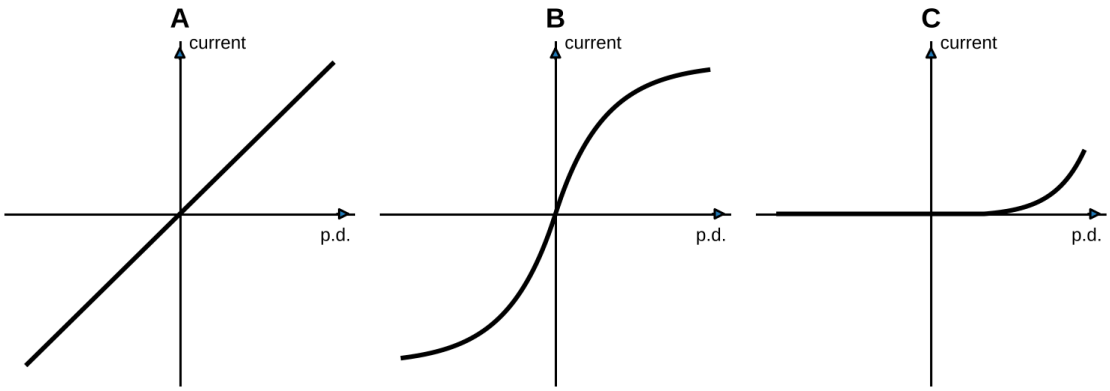
0 3 . 5 State what happens to the current if one of the resistors is removed from the circuit and the other resistor is left in place. **[2 marks]**

_____

_____

_____

0 4 The graphs show the current–potential difference characteristics of three components, **A**, **B** and **C**.



0 4 . 1 Name the three components. **[3 marks]**

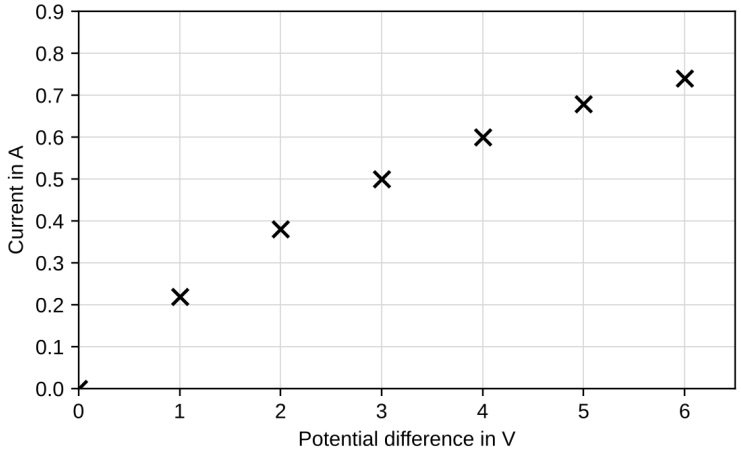
0 4 . 2 Explain why the graph for component **A** is a straight line through the origin. **[2 marks]**

0 4 . 3 Explain why the resistance of component **B** increases as the p.d. increases. **[3 marks]**

0 4 . 4 Describe the behaviour of component **C** when the p.d. is reversed. **[2 marks]**

0 4 . 5 Give **one** use of a diode. **[1 mark]**

0 5 A student investigates how the current through a filament lamp depends on the potential difference across it. The graph shows her results.



0 5 . 1 Draw a circuit diagram the student could use. Include a cell, a variable resistor, an ammeter, a voltmeter and the lamp. **[4 marks]**

0 5 . 2 Draw a smooth curve of best fit through the points. **[1 mark]**

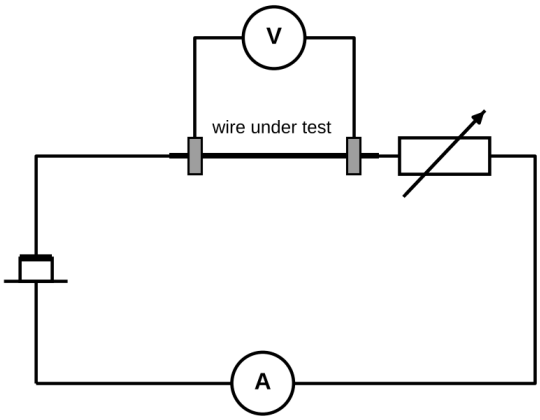
0 5 . 3 Calculate the resistance of the lamp when the potential difference is 4.0 V. **[3 marks]**

Answer = _____ Ω

0 5 . 4 The resistance at 1.0 V is smaller than the resistance at 4.0 V. Calculate the resistance at 1.0 V and explain the difference. **[4 marks]**

0 5 . 5 Suggest why the student reverses the cell connections and repeats the experiment. **[1 mark]**

0 6 A student investigates how the resistance of a wire depends on its length. The diagram shows the circuit. The potential difference across the wire is 1.5 V. The table shows her results.



0 6 . 1 State **one** variable the student should control. **[1 mark]**

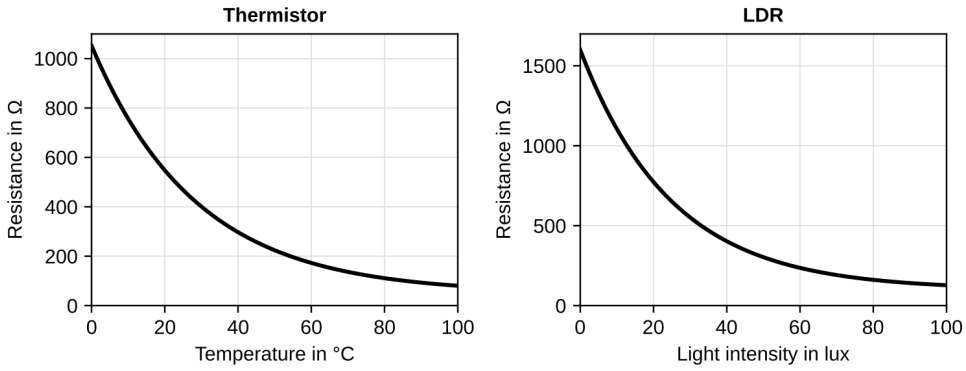
0 6 . 2 Complete the table by calculating the resistance for each length. **[4 marks]**

Length in cm	Current in A	Resistance in Ω
20	0.75	
40	0.38	
60	0.25	
80	0.19	
100	0.15	

0 6 . 3 Describe the relationship between the length of the wire and its resistance. **[2 marks]**

0 6 . 4 Explain why the student switches off the current between readings. **[2 marks]**

0 7 The graphs show how the resistance of a thermistor and of a light-dependent resistor (LDR) change.



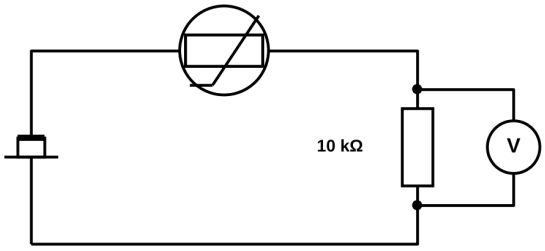
0 7 . 1 Use the graph to find the resistance of the thermistor at 20 °C. **[1 mark]**

0 7 . 2 Describe how the resistance of the LDR changes as the light intensity increases. **[1 mark]**

0 7 . 3 Give a use for a thermistor. **[1 mark]**

0 7 . 4 Give a use for an LDR. **[1 mark]**

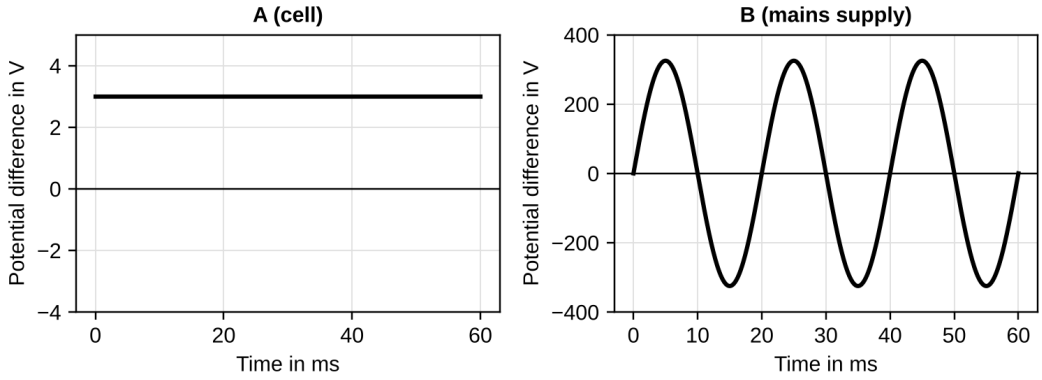
0 8 A thermistor is connected in series with a 10 kΩ resistor and a cell. A voltmeter is connected across the 10 kΩ resistor.



0 8 . 1 The temperature of the thermistor rises. Explain what happens to the reading on the voltmeter. **[3 marks]**

0 8 . 2 The potential difference across the thermistor is the cell p.d. minus the p.d. across the resistor. Explain why. **[2 marks]**

0 9 The graphs show the potential difference from two sources, **A** and **B**.



0 9 . 1 Which graph shows a direct potential difference (d.c.)? **[1 mark]**

0 9 . 2 Give **one** difference between direct and alternating potential difference. **[1 mark]**

0 9 . 3 State the frequency of the mains supply in the UK. **[1 mark]**

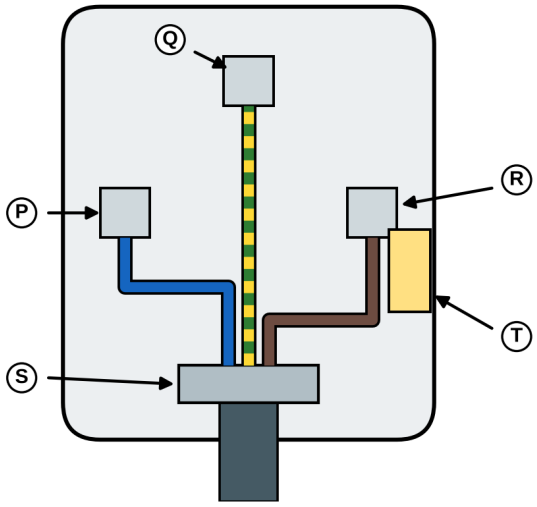
0 9 . 4 Use graph **B** to find the time for one complete cycle. **[1 mark]**

0 9 . 5 Calculate the frequency of supply B using your answer. **[3 marks]**

Answer = _____ Hz

0 9 . 6 State the value of the mains potential difference (rms) in the UK. **[1 mark]**

1 0 The diagram shows the inside of a three-pin mains plug.



1 0 . 1 Name the wires **P**, **Q** and **R** and give the colour of the insulation on each. **[6 marks]**

1 0 . 2 Name the parts labelled **S** and **T**. **[2 marks]**

1 0 . 3 State the potential difference between the live wire and earth. **[1 mark]**

1 0 . 4 Explain how the earth wire and the fuse work together to protect the user if the live wire touches the metal case of an appliance. **[4 marks]**

1 0 . 5 A fault means a hairdryer draws a current of 15 A. The plug contains a 13 A fuse. State and explain what happens. **[2 marks]**

1 1 This question is about electrical power.

1 1 . 1 Write down the **two** equations that link power with other quantities (not involving energy). **[2 marks]**

1 1 . 2 An electric kettle is connected to the 230 V mains. It has a power of 2300 W. Calculate the current in the kettle. **[3 marks]**

Answer = _____ A

1 1 . 3 Calculate the resistance of the kettle element. **[3 marks]**

Answer = _____ Ω

1 1 . 4 A heater element has a resistance of 8.0 Ω and a current of 3.0 A flows through it. Calculate the power. **[3 marks]**

Answer = _____ W

1 1 . 5 Which fuse should be fitted to the kettle: 3 A, 5 A or 13 A? Give a reason. **[2 marks]**

_____

_____

_____

1 1 . 6 A 2300 W kettle is used for 3.0 minutes. Calculate the energy transferred in kWh. **[3 marks]**

Answer = _____ kWh

1 1 . 7 Energy costs 28 p per kWh. Calculate the cost of the energy in part 6. **[2 marks]**

Answer = _____ p

1 2 This question is about energy transfers in circuits.

1 2 . 1 Write down the equation that links energy transferred, charge flow and potential difference. **[1 mark]**

1 2 . 2 A 12 V battery moves 5.0 C of charge. Calculate the energy transferred. **[3 marks]**

Answer = _____ J

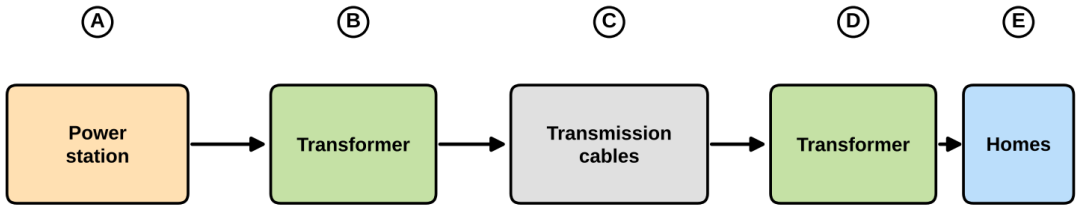
1 2 . 3 A hairdryer has a power of 1500 W. Calculate the energy transferred in 4.0 minutes. **[3 marks]**

Answer = _____ J

1 2 . 4 Explain why a current in a wire transfers energy to thermal stores. **[3 marks]**

1 2 . 5 Give **one** advantage and **one** disadvantage of the heating effect of a current. **[2 marks]**

1 3 The diagram shows how electricity is transmitted from a power station to homes by the National Grid.



1 3 . 1 Name the type of device at **B** and **D**. **[2 marks]**

1 3 . 2 Explain why electricity is transmitted at a very high potential difference. **[4 marks]**

1 3 . 3 Why is the potential difference reduced before it reaches homes? **[1 mark]**

1 3 . 4 A power station supplies 100 MW at 400 000 V. Calculate the current in the cables. **[3 marks]**

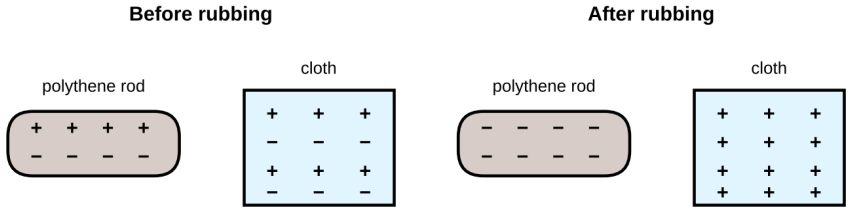
Answer = _____ A

1 3 . 5 The cables have a resistance of 10 Ω. Calculate the power wasted in the cables. **[3 marks]**

Answer = _____ W

1 3 . 6 If the power was transmitted at 25 000 V instead, the power wasted would be 160 MW. Explain why this would not work. **[2 marks]**

1 4 A student rubs a polythene rod with a dry cloth. The rod becomes negatively charged.



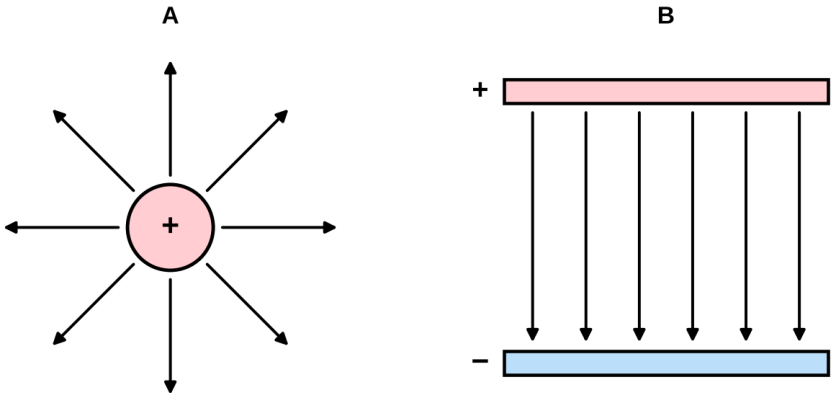
1 4 . 1 Explain how the rod becomes negatively charged. **[3 marks]**

1 4 . 2 State the charge on the cloth. **[1 mark]**

1 4 . 3 The student holds the charged rod near a second charged rod. They repel. State the charge on the second rod. **[1 mark]**

1 4 . 4 State what is meant by 'like charges repel and opposite charges attract'. Give **one** example of a non-contact force from this idea. **[2 marks]**

1 5 The diagrams show the electric fields around a positively charged sphere (**A**) and between two parallel plates (**B**).



1 5 . 1 What is an electric field? **[1 mark]**

1 5 . 2 In diagram **A**, what does the direction of the arrows show? **[1 mark]**

1 5 . 3 A small negative charge is placed in the field in diagram **B**. State the direction of the force on it. **[1 mark]**

1 5 . 4 State how the diagram shows that the field is stronger close to the sphere in **A**. **[1 mark]**

1 5 . 5 A charged plastic tanker is refuelled. Explain why a metal wire is connected between the tanker and the ground. **[2 marks]**

1 6 Read the question and answer in full sentences.

1 6 . 1 *Describe an investigation to find out how the current through a resistor depends on the potential difference across it. Your answer should include a circuit diagram description, the measurements taken and how the results are analysed. **[6 marks]**

1 6 . 2 *A student says: 'All mains appliances in my house are safe because there is a fuse in the plug.' Evaluate this statement. Refer to fuses, earth wires and double insulation. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.2: Electricity

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	ammeter		1	AO1 4.2.1.1
01.2	in parallel (with the lamp)		1	AO1 4.2.1.1
01.3	charge flow = current × time		1	AO1 4.2.1.2
01.4	t = 120 s ; Q = 0.40 × 120 ; 48 (C) ;	Common error: using 2.0 instead of 120 s (0.8 C, 2 marks max).	3	AO2 4.2.1.2
01.5	R = V ÷ I ; = 3.0 ÷ 0.40 ; 7.5 (Ω) ;		3	AO2 4.2.1.3
01.6	E = Q × V ; = 48 × 3.0 ; 144 (J) ;	Allow ecf from part 4. Allow P × t = 1.2 × 120.	3	AO2 4.2.1.2
01.7	decreases / halves ;		1	AO2 4.2.3
02.1	I = V ÷ R ; = 12 ÷ 6.0 ; 2.0 (A) ;		3	AO2 4.2.1.3
02.2	current in the main circuit = sum of currents in branches ; 2.0 + 1.0 = 3.0 (A) ;		2	AO2 4.2.3
02.3	R = V ÷ I ; 12 ÷ 1.0 ; 12 (Ω) ;		3	AO2 4.2.1.3
02.4	R = V ÷ I ; = 12 ÷ 3.0 ; 4.0 (Ω) ;	Allow ecf from part 2.	3	AO2 4.2.3
02.5	adding a resistor in parallel gives the current another path / more branches ; so more current flows from the battery for the same p.d. / total current increases ;		2	AO3 4.2.3
02.6	12 (V)	Same p.d. across each branch in a parallel circuit.	1	AO2 4.2.3
03.1	4.0 + 6.0 ; 10 (Ω) ;		2	AO2 4.2.3
03.2	I = V ÷ R ; = 5.0 ÷ 10 ; 0.50 (A) ;		3	AO2 4.2.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	V = I × R ; = 0.50 × 6.0 ; 3.0 (V) ;	Allow ecf from part 2.	3	AO2 4.2.3
03.4	2.0 (V)	Allow 5.0 − 3.0.	1	AO2 4.2.3
03.5	current increases ; because total resistance decreases ;		2	AO2 4.2.3
04.1	A: resistor (at constant temperature) ; B: filament lamp ; C: diode ;	Allow ohmic conductor for A.	3	AO1 4.2.1.4
04.2	current is directly proportional to p.d. ; resistance is constant (as long as temperature is constant) ;		2	AO1 4.2.1.4
04.3	greater current causes more heating (of the filament) ; ions / atoms vibrate more ; more collisions with electrons so greater resistance ;		3	AO2 4.2.1.4
04.4	no current / (almost) zero current flows ; diode only allows current in one direction ;		2	AO1 4.2.1.4
04.5	rectifier / converts ac to dc / protect circuit if cell is connected backwards		1	AO1 4.2.1.4
05.1	lamp, cell, ammeter and variable resistor in series ; voltmeter in parallel with the lamp ; correct symbols for all components ; complete circuit with no gaps ;		4	AO1 4.2.1.4 RP
05.2	smooth curve through the points ignoring anomalies (and passing through the origin) ;		1	AO3 WS 3.3
05.3	I = 0.60 A (from graph) ; R = 4.0 ÷ 0.60 ; 6.7 (Ω) ;	Allow 6.67.	3	AO3 4.2.1.3
05.4	R = 1.0 ÷ 0.22 ; 4.5 (Ω) ; at higher p.d. the filament is hotter ; so the resistance increases ;		4	AO3 4.2.1.4
05.5	to collect results for negative values of p.d. / check the symmetry of the characteristic		1	AO3 4.2.1.4 RP
06.1	thickness / diameter / material / temperature of the wire (keep the p.d. constant)		1	AO3 4.2.1.3 RP
06.2	R = V ÷ I used ; 2.0 ; 3.9 / 4.0 ; 6.0, 7.9 / 8.0 and 10 ;	Resistance values: 2.0, 3.9 (allow 4.0), 6.0, 7.9 (allow 8.0), 10 (Ω). Award marks for method + any 3 correct values (3), all correct (4).	4	AO2 4.2.1.3 RP
06.3	as length increases, resistance increases ; directly proportional / doubling the length doubles the resistance ;		2	AO3 4.2.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	to stop the wire heating up ; heating would increase the resistance / affect the results ;		2	AO3 4.2.1.3 RP
07.1	530–570 (Ω)		1	AO3 WS 3.1
07.2	decreases ;		1	AO3 4.2.1.5
07.3	thermostat / temperature sensor / fire alarm / (heating) control		1	AO1 4.2.1.5
07.4	automatic street lights / security lights / light sensor		1	AO1 4.2.1.5
08.1	resistance of the thermistor decreases ; current in the circuit increases ; so the p.d. across the fixed resistor (voltmeter reading) increases ;		3	AO3 4.2.1.5
08.2	components are in series ; p.d. of the cell is shared between components (total = sum of parts) ;		2	AO2 4.2.3
09.1	A		1	AO3 4.2.2
09.2	alternating p.d. changes direction (repeatedly); direct p.d. is always in the same direction		1	AO1 4.2.2
09.3	50 (Hz)		1	AO1 4.2.2.1
09.4	20 (ms) / 0.020 (s)		1	AO3 WS 3.1
09.5	f = 1 ÷ T ; 1 ÷ 0.020 ; 50 (Hz) ;		3	AO2 4.2.2
09.6	230 (V)		1	AO1 4.2.2.1
10.1	P: neutral ; blue ; Q: earth ; green and yellow (stripes) ; R: live ; brown ;		6	AO1 4.2.2.2
10.2	S: cable grip ; T: fuse ;		2	AO1 4.2.2.2
10.3	230 (V)		1	AO1 4.2.2.2
10.4	the earth wire connects the metal case to earth ; a large current flows in the earth wire ; the fuse melts / breaks the circuit ; so the case is not live / user does not get a shock ;		4	AO2 4.2.2.2
10.5	the fuse melts / blows ; current is greater than the fuse rating so the circuit is broken ;		2	AO2 4.2.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
11.1	power = potential difference × current ; power = (current)^2 × resistance ;		2	AO1 4.2.4.1
11.2	I = P ÷ V ; = 2300 ÷ 230 ; 10 (A) ;		3	AO2 4.2.4.1
11.3	R = V ÷ I ; = 230 ÷ 10 ; 23 (Ω) ;	Allow R = P ÷ I^2 = 2300 ÷ 100.	3	AO2 4.2.4.1
11.4	P = I^2 R ; = 3.0^2 × 8.0 ; 72 (W) ;		3	AO2 4.2.4.1
11.5	13 A ; the normal current (10 A) is below 13 A but above 3 A and 5 A, so those fuses would blow ;		2	AO3 4.2.2.2
11.6	2.3 kW and 0.050 h ; E = P × t = 2.3 × 0.050 ; 0.115 (kWh) ;	Allow 0.12. Alternatively 2300 × 180 = 414 000 J.	3	AO2 4.2.4.2
11.7	0.115 × 28 ; 3.2 p ;	Allow ecf. Allow 3.22 p.	2	AO2 4.2.4.2
12.1	energy transferred = charge flow × potential difference		1	AO1 4.2.4.2
12.2	E = Q × V ; = 5.0 × 12 ; 60 (J) ;		3	AO2 4.2.4.2
12.3	t = 240 s ; E = 1500 × 240 ; 360 000 (J) ;		3	AO2 4.2.4.1
12.4	electrons collide with ions in the wire ; ions vibrate more ; so the wire gets hot / thermal store increases ;		3	AO1 4.2.4.1
12.5	advantage: heating (kettle, toaster, heater) ; disadvantage: wasted energy in wires / risk of fire ;		2	AO1 4.2.4.1
13.1	B: step-up transformer ; D: step-down transformer ;		2	AO1 4.2.5
13.2	for the same power, increasing p.d. decreases current ; P = V I ; less energy is wasted as heat in the cables ; since power loss = I^{2}R ;		4	AO2 4.2.5
13.3	for safety / appliances need 230 V		1	AO1 4.2.5
13.4	I = P ÷ V ; = 100 000 000 ÷ 400 000 ; 250 (A) ;		3	AO2 4.2.5
13.5	P = I^2 R ; = 250^2 × 10 ; 625 000 (W) / 0.625 MW ;		3	AO2 4.2.5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
13.6	160 MW is greater than the 100 MW supplied ; so no useful power / more energy wasted than generated ;		2	AO3 4.2.5
14.1	friction (rubbing) transfers electrons ; electrons move from the cloth to the rod ; the rod gains a negative charge (only electrons move) ;		3	AO2 4.2.6
14.2	positive	Allow equal and opposite to the rod.	1	AO2 4.2.6
14.3	negative		1	AO2 4.2.6
14.4	same charges push apart, different charges pull together ; e.g. charged rod attracting paper / balloon sticking to wall ;		2	AO1 4.2.6
15.1	a region where a charged object experiences a force		1	AO1 4.2.6
15.2	the direction of the force on a (small) positive charge		1	AO1 4.2.6
15.3	towards the positive plate / upwards		1	AO2 4.2.6
15.4	field lines are closer together (near the sphere)		1	AO2 4.2.6
15.5	charge flows to earth / stops build-up of charge ; prevents sparks that could ignite the fuel ;		2	AO2 4.2.6
16.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6 marks): complete, logical method with a correct circuit, measurements and analysis including a graph and conclusion. • Level 2 (3–4 marks): mostly correct method; some detail missing. • Level 1 (1–2 marks): simple statements. • Indicative content: ammeter in series, voltmeter in parallel with the resistor; variable resistor / change the number of cells to vary p.d.; record p.d. and current for at least five values; keep the temperature constant (switch off between readings); repeat and take means; plot current against p.d.; straight line through origin shows current ∝ p.d.; gradient = 1 ÷ R.		6	AO3 4.2.1.4 RP

Question	Answers	Extra information	Mark	AO / Spec. Ref.
16.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6 marks): a balanced evaluation of at least two safety features, with a clear judgement. • Level 2 (3–4): some advantages and limitations. • Level 1 (1–2): simple statements. • Indicative content: fuse melts when current too high; breaks circuit; but it only works if the fault current exceeds the fuse rating; wrong fuse (e.g. 13 A in a 3 A appliance) is unsafe; earth wire needed for metal-cased appliances to cause a large current; double-insulated appliances (plastic cases) need no earth; circuit breakers / RCDs switch off faster; fuse must be in the live wire; judgement that the fuse is helpful but not enough on its own.		6	AO3 4.2.2.2

Where marks are lost in Topic 4.2

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Series vs parallel:** series – same current, p.d. shared, resistances add. Parallel – same p.d., currents add, total resistance is less than the smallest resistor.
- ☐ **Ammeters in series, voltmeters in parallel.** Draw them with a circle containing A or V. Never leave gaps in the circuit.
- ☐ **Equations:** $V = I\,R$, $Q = I\,t$, $E = Q\,V$, $P = V\,I = I^2\,R$, $E = P\,t$. Write the equation, substitute, then answer with the unit.
- ☐ **I–V graphs:** resistor – straight line through the origin; filament lamp – curve that gets flatter (resistance rises with temperature); diode – current in one direction only.
- ☐ **Explain a higher resistance** using: more heating → ions vibrate more → more collisions with electrons.
- ☐ **Plug wires:** blue = neutral, brown = live (230 V), green/yellow = earth. Fuse goes in the live wire.
- ☐ **National Grid:** step-up transformer → high p.d., low current → less energy wasted as heat → step-down transformer. Say **why** using $P = VI$ and $P = I^2R$.
- ☐ **Units:** convert minutes to seconds (× 60) and kW to W (× 1000). For kWh use kW × hours.
- ☐ **Static:** only **electrons** move. The object that gains electrons becomes negative. Never say 'positive charges moved'.